WCCM-ECCOMAS 2026 | MS165 - METHODS AND APPLICATIONS OF MODEL ORDER REDUCTION

Hybrid Green's Function Learning With Axial Reduction

A two-dimensional elliptic solver built from axial Green operators and a learned directional source split

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REDUCE
a multi-dimensional elliptic problem into axial Green subproblems

COUPLE
learn the directional source split that makes axial reconstructions consistent

RECOVER
a multi-dimensional solution from coupled axial Green operators

Axial Reduction of Elliptic Operators

Replace a global source-to-solution map with structured one-dimensional Green reconstructions

$$\begin{aligned} -\nabla\cdot(a\nabla u) + \mathbf{b}\cdot\nabla u + cu &= f \quad \text{in } \Omega, \\ u &= 0 \quad \text{on } \partial\Omega, \end{aligned} \qquad \mathbf{b} = (b_x, b_y).$$

$$\mathcal{S}_{a,\mathbf{b},c} : f \mapsto u, \qquad u = \mathcal{S}_{a,\mathbf{b},c}[f].$$

Operator fixed; source varies.

DIRECT MAP

one global source-to-solution representation

High-dimensional and geometry-sensitive

→

REDUCED VIEW

directional 1-D operators + learned coupling

line-wise inverses + learned source split

$$L_x u = -\partial_x(a\partial_x u) + b_x\partial_x u + \frac{1}{2}cu, \qquad L_y u = -\partial_y(a\partial_y u) + b_y\partial_y u + \frac{1}{2}cu.$$

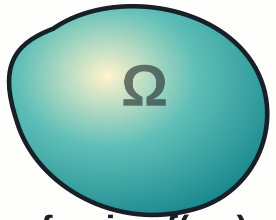
$$L_x u + L_y u = f.$$

GreenNet supplies line-wise Green inverses; CouplingNet learns the source split that turns them into a 2D elliptic solver.

From a 2D Elliptic Problem to Coupled Axial Green Solves

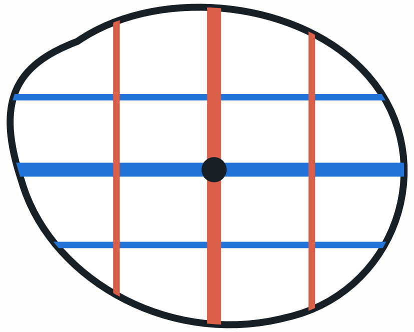
Graphic abstract: slice, solve line-wise, learn the coupling, reconstruct the field

2D FORCING PROBLEM



forcing f(x,y)

AXIAL INTERVAL VIEW



axial intervals inside Ω

GREENNET: LINE-WISE INVERSE

KERNEL INTEGRATION

$$v(t) = \int_0^1 G_\theta(t,\eta)\rho(\eta)\,d\eta$$

line source profile -> line solution

COUPLINGNET: DIRECTIONAL COUPLING

SOURCE SPLIT COUPLES LINE-WISE INVERSES

$$f \xrightarrow{\text{CouplingNet}} (\phi,\psi)$$

$$\phi + \psi = f$$

$$\phi \xrightarrow{G_x} u_\phi, \qquad \psi \xrightarrow{G_y} u_\psi \Rightarrow u$$

Axial GreenNet provides line-wise inverses; CouplingNet learns the directional source split that couples them into a 2D solution.

Unit-Interval Pull-Back and Operator Scaling

Physical length is absorbed into normalized operator coefficient profiles and source profile

PHYSICAL ONE-DIMENSIONAL AXIAL OPERATOR

$$s \in [s_0, s_1],$$
$$\mathcal{L}_{\mathrm{phys}} u = -\frac{d}{ds}\left(a_{\mathrm{phys}}(s)\frac{du}{ds}\right) + b_{\parallel,\mathrm{phys}}(s)\frac{du}{ds} + c_{\mathrm{phys}}(s)u = f_{\mathrm{phys}}(s).$$

NORMALIZE THE COORDINATE

$$s = s_0 + Lt, \qquad v(t) = u(s_0 + Lt), \qquad t \in [0,1].$$

The interval is normalized, but the physical length is not discarded.

$$a_{\mathrm{unit}} = a_{\mathrm{phys}}, \quad a'_{\mathrm{unit}} = La'_{\mathrm{phys}}, \quad b_{\parallel,\mathrm{unit}} = Lb_{\parallel,\mathrm{phys}}, \quad c_{\mathrm{unit}} = L^2 c_{\mathrm{phys}}, \quad f_{\mathrm{unit}} = L^2 f_{\mathrm{phys}}.$$

$$-\frac{d}{dt}\left(a_{\mathrm{unit}}(t)\frac{dv}{dt}\right) + b_{\parallel,\mathrm{unit}}(t)\frac{dv}{dt} + c_{\mathrm{unit}}(t)v(t) = f_{\mathrm{unit}}(t), \qquad t \in [0,1].$$

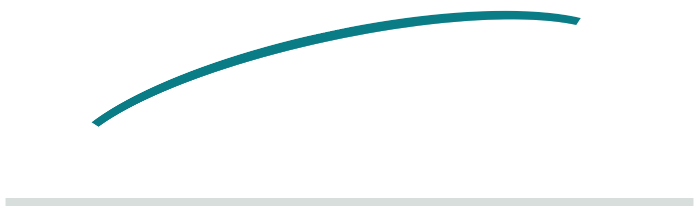
Here $b_\parallel = b_x$ on an x-directed interval and $b_\parallel = b_y$ on a y-directed interval.

GreenNet I: Normalized Axial Green Operator

Learn one-dimensional Green kernels, not a global two-dimensional kernel

Previous step: normalize the operator. **This step: learn its Green inverse.**

SOURCE PROFILE



$f_{\text{unit}}(\eta)$

$G_{\text{unit}}(t,\eta)$

KERNEL INTEGRAL OPERATOR

SOLUTION PROFILE

$v(t)$

$$\mathcal{G}_{\text{unit}}[f_{\text{unit}}](t) = \int_0^1 G_{\text{unit}}(t,\eta) f_{\text{unit}}(\eta)\, d\eta, \qquad v = \mathcal{G}_{\text{unit}}[f_{\text{unit}}].$$

Operator coefficients: Axial coefficient profiles define each local 1D operator.

Kernel coordinates (t,η) define the evaluation and source points.

GreenNet target is a local unit-interval kernel, not a global 2D Green function.

GreenNet II: Analytic Green Structure and Learned Correction

A hybrid kernel: Dirac delta structure, Heaviside cancellation, and learned smooth correction

Do not learn the Green singularity from scratch.

HYBRID GREEN KERNEL

$$G_\theta = \text{Dirac delta structure} + \text{Heaviside cancellation} + \text{learned smooth correction.}$$

DIRAC DELTA STRUCTURE

Build the source jump.

HEAVISIDE CANCELLATION

Cancel the induced Heaviside contribution.

LEARNED SMOOTH CORRECTION

Learn the smooth residual.

Analytic structure handles the singular terms before learning.

GreenNet II: Analytic Green Structure and Learned Correction

A hybrid kernel: Dirac delta structure, Heaviside cancellation, and learned smooth correction

$$G_\theta(t,\eta) = \underbrace{E(t,\eta)M(t)R_\theta(t,\eta)}_{\text{learned smooth correction}} + \underbrace{B(t)\left(J_0(t,\eta) - \frac{1}{2}E(t,\eta)\right)}_{\text{Heaviside cancellation}} + \underbrace{A(t)G_0(t,\eta)}_{\text{Dirac delta jump}}.$$

DIRAC DELTA STRUCTURE

$$G_0(t,\eta) = \begin{cases} t(1-\eta), & t<\eta, \\ \eta(1-t), & t \geq \eta, \end{cases} \qquad \partial_t^2 G_0 = -\delta(t-\eta).$$

HEAVISIDE CANCELLATION

$$\partial_t J_0(t,\eta) = G_0(t,\eta), \quad S = J_0 - \frac{1}{2}E, \quad \partial_t^2 S = \partial_t G_0.$$

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LEARNED SMOOTH CORRECTION

$$E(t,\eta)M(t)R_\theta(t,\eta), \qquad E(t,\eta) = t\eta(1-\eta), \qquad M(t) = 1-t.$$

GreenNet learns R_θ; $E(t,\eta)M(t)$ keeps that correction boundary-compatible. The singular and cancellation terms are already built into the kernel form.

The neural network learns the smooth residual with boundary-compatible envelope factors.

GreenNet III: Source-to-Solution Supervision

Supervise the Green operator action, not pointwise exact-kernel labels

No pointwise Green-kernel labels.

$w(t) \sim \mathcal{GP}(0, k_\ell)$ → $v(t) = w(t) - ((1-t)w(0) + t\,w(1))$ so $v(0) = v(1) = 0$ → $f_{\text{unit}} = \mathcal{L}_{\text{unit}} v$

$$v_\theta(t) = \int_0^1 G_\theta(t,\eta) f_{\text{unit}}(\eta)\, d\eta.$$

$$\mathcal{J}_{\text{Green}} \sim \mathbb{E}\left[\int_0^1 |v_\theta(t) - v(t)|^2 \; dt\right].$$

Training checks whether the learned Green operator maps sources generated from target solutions back to those target solutions.

Numerical Evidence I: GreenNet Kernel Approximation

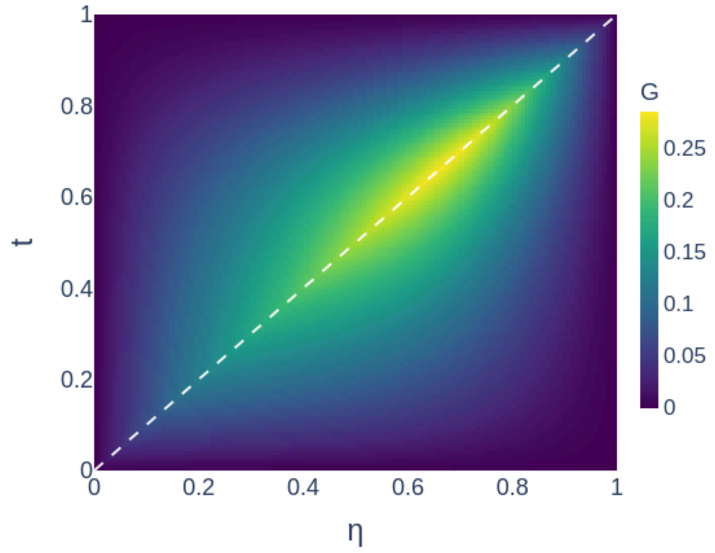
Capturing the singular Green structure on an axial interval

$$\Omega = \{x^2 + y^2 < 0.5^2\}, \quad -\nabla \cdot (a \nabla u) + \mathbf{b} \cdot \nabla u = f, \quad c = 0,$$

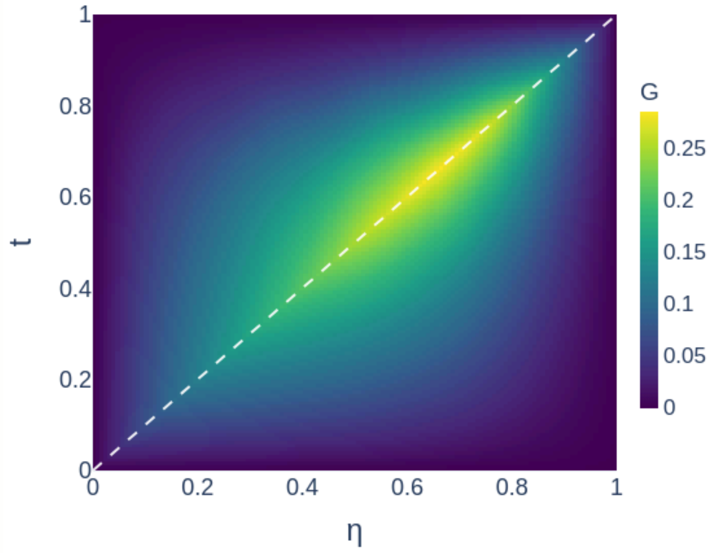
$$a(x,y) = 1 + \frac{1}{2}\sin(2\pi x)\sin(2\pi y), \quad \mathbf{b}(x,y) = \left(\frac{1}{2}\sin(\pi x)\cos(\pi y), -\frac{1}{2}\cos(\pi x)\sin(\pi y)\right).$$

Axial line: y-directed interval at x = -0.25, L = 0.866.

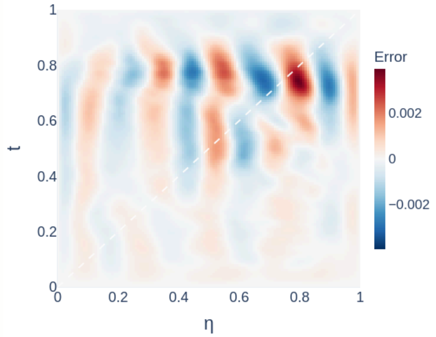
REFERENCE KERNEL



LEARNED KERNEL



SIGNED ERROR



DIAGNOSTICS

η	0.75
Kernel rel. L2	0.65%
Slice rel. L2	0.46%
Diagonal band	|t-η| ≤ 5/128 = 0.0391
Diagonal error mass / area	11.8% / 8.3%
Diagonal mean / off-band mean	1.47x

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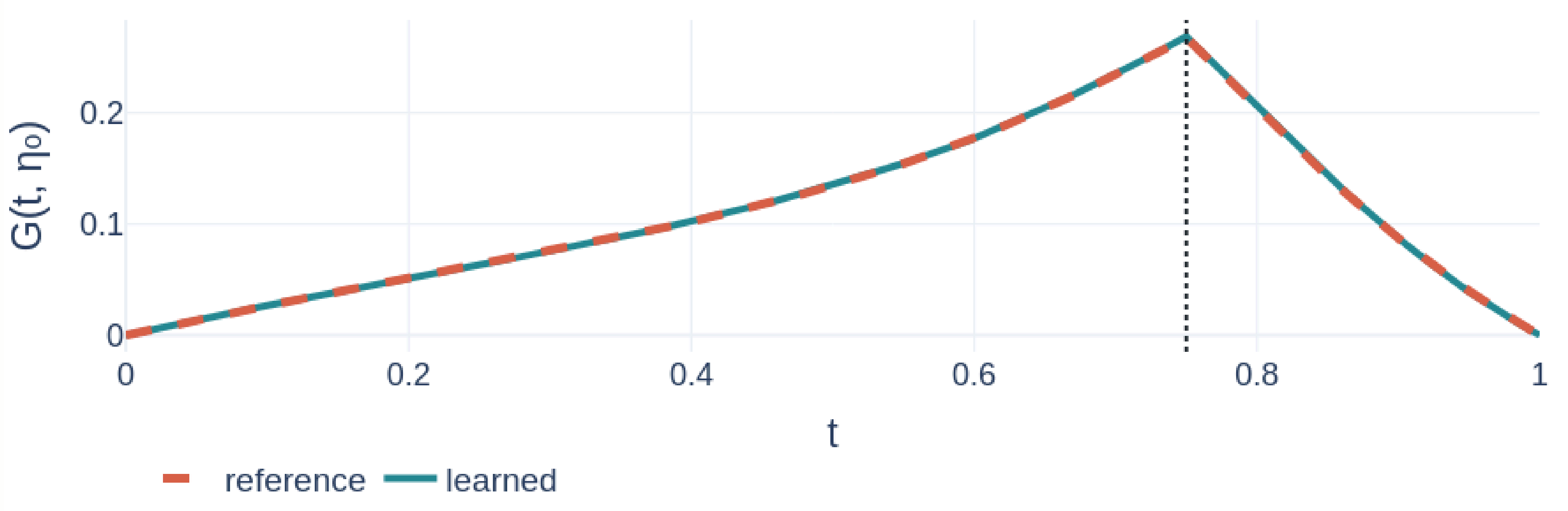
FIXED-η SLICE, η=0.75



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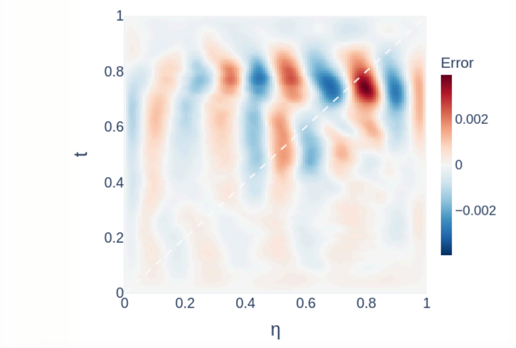

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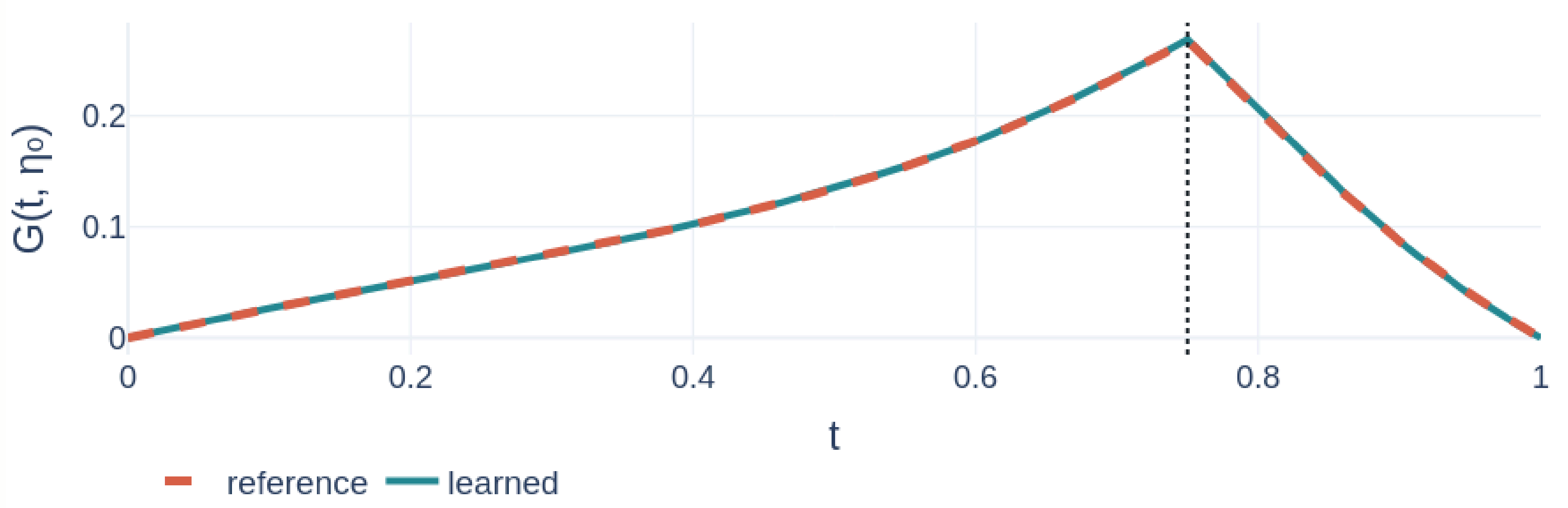

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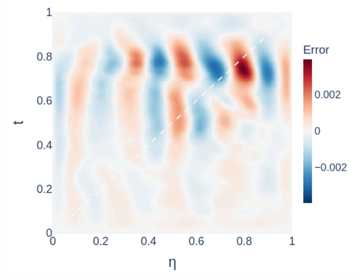

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The learned kernel captures the singular Green structure, and the signed error is not concentrated near the singular diagonal.

CouplingNet I: Directional Source Split

Learn how the forcing is divided between axial Green reconstructions

$G_x[\cdot]$

full source f

?

$G_y[\cdot]$

$$f \longmapsto (\phi, \psi).$$

X-DIRECTED SOURCE

$$\phi = L_x u, \qquad L_x u = -\partial_x(a\partial_x u) + b_x\partial_x u + \tfrac{1}{2}cu.$$

Y-DIRECTED SOURCE

$$\psi = L_y u, \qquad L_y u = -\partial_y(a\partial_y u) + b_y\partial_y u + \tfrac{1}{2}cu.$$

$$\phi + \psi = f.$$

CouplingNet learns line-wise flux-divergence/source components that couple the axial Green reconstructions.

CouplingNet II: Branches and Local Context for Split Prediction

Branch nets encode profiles; trunk nets encode pointwise coordinates

BRANCH NETS

SOURCE PROFILE

sample-varying forcing profile

COEFFICIENT PROFILES

diffusion, primary/transverse convection, reaction

LINE-GEOMETRY STRUCTURE

interval length, endpoints, transverse placement

SPLIT PREDICTOR

$$(\text{branch features},\ \text{trunk features}) \longrightarrow (\phi, \psi)$$

TRUNK NETS

AXIAL LOCAL TRUNK

$t_{\parallel}$: position along the primary interval

POINTWISE TRANSVERSE TRUNK

$t_{\perp}$: position on the transverse interval through this point

$$\underbrace{\text{profiles and line geometry}}_{\text{branch nets}} + \underbrace{\text{pointwise coordinates}}_{\text{trunk nets}} \Rightarrow \text{directional source split.}$$

CouplingNet III: Projection and Green Reconstruction

Project in physical split variables, then reconstruct and average

STAGE 1: PHYSICAL BALANCE PROJECTION

COUPLINGNET RAW SPLIT

$$(\phi_{\text{raw}}, \psi_{\text{raw}})$$

not guaranteed balanced

→

PROJECTION STEP

$$r = f - (\phi_{\text{raw}} + \psi_{\text{raw}})$$

$$\phi = \phi_{\text{raw}} + \frac{1}{2}r, \qquad \psi = \psi_{\text{raw}} + \frac{1}{2}r$$

→

BALANCED SPLIT

$$\phi + \psi = f$$

STAGE 2: GREEN RECONSTRUCTION

X-REPRESENTED SOLUTION

$$u_\phi = G_x[\phi]$$

Y-REPRESENTED SOLUTION

$$u_\psi = G_y[\psi]$$

FINAL AVERAGE

$$u_{\text{pred}} = \frac{1}{2}(u_\phi + u_\psi)$$

$$(\phi_{\text{raw}}, \psi_{\text{raw}}) \xrightarrow{\text{projection}} (\phi, \psi),\ \phi + \psi = f \xrightarrow{G_x, G_y} (u_\phi, u_\psi) \xrightarrow{\text{average}} u_{\text{pred}}.$$

Energy-Norm Error Bound Proposition

Small split energy controls the final energy error

u_ϕ

$u_{\text{pred}} = \frac{1}{2}(u_\phi + u_\psi)$

u_ψ

DIFFUSION-WEIGHTED ENERGY NORM

$$\|v\|_a^2 := \int_\Omega a(x)|\nabla v(x)|^2\,dx, \qquad a(x) > 0.$$

Here $a(x)$ is the diffusion coefficient.

SPLIT-ENERGY LOSS

$$\mathcal{E}_{\text{split}} = \|u_\phi - u_\psi\|_a^2.$$

REFERENCE SOLUTION

$$\mathcal{L}u_* = f, \qquad u_*|_{\partial\Omega} = 0.$$

$$\|u_{\text{pred}} - u_*\|_a \le \frac{C_E}{2}\sqrt{\mathcal{E}_{\text{split}}}.$$

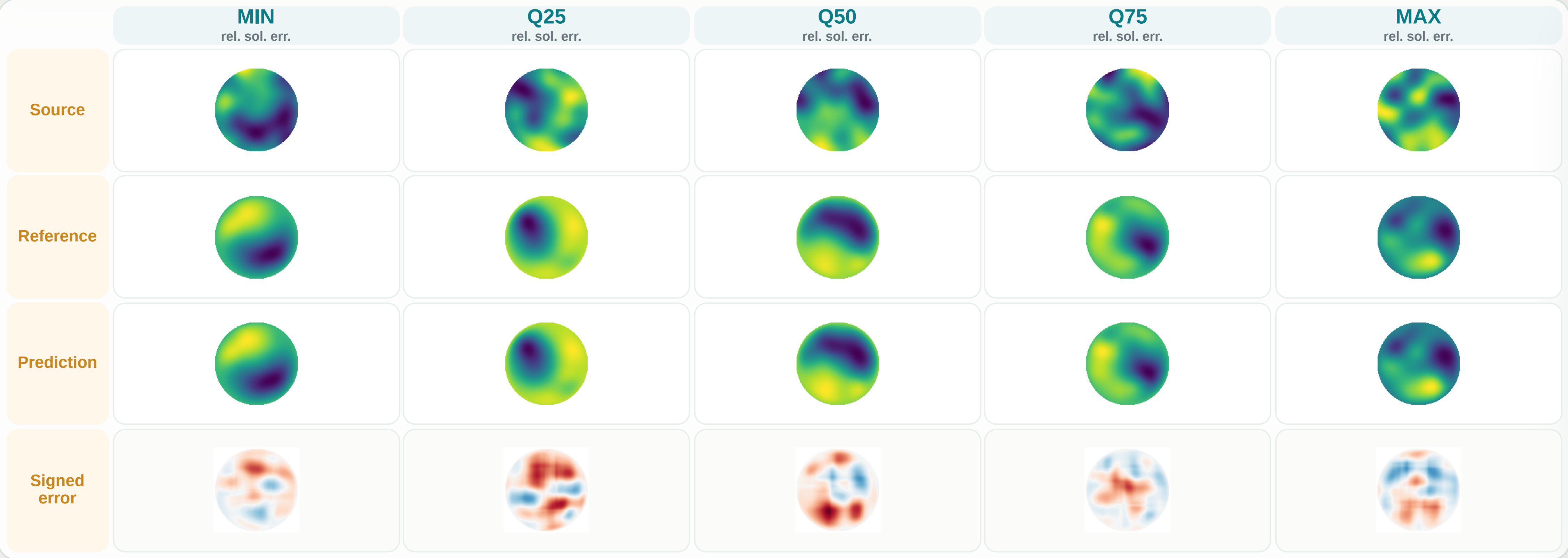
C_E: stability constant of the fixed elliptic operator.

Conditional on exact/controlled Green reconstruction and $H_0^1(\Omega)$-admissible represented solutions.

Numerical Evidence II: CouplingNet Solution Reconstruction

Quantile-selected samples show source, reference, prediction, and signed error

$$\Omega = \{x^2 + y^2 < 0.5^2\}, \qquad -\nabla\cdot(a\nabla u) + \mathbf{b}\cdot\nabla u + cu = f,$$
$$a = 1 + \frac{1}{2}\sin(2\pi x)\sin(2\pi y), \qquad \mathbf{b} = \left(\frac{1}{2}\sin(\pi x)\cos(\pi y), -\frac{1}{2}\cos(\pi x)\sin(\pi y)\right), \quad c = \frac{1}{2}\left(1 + \frac{1}{2}\cos(2\pi x)\cos(2\pi y)\right).$$



SELECTED BY RELATIVE SOLUTION ERROR

CASE	REL. SOL. ERR.	SPLIT ENERGY
min	2.33%	3.04e-4
q25	2.74%	5.10e-4
q50	3.32%	4.59e-4
q75	4.32%	2.51e-4
max	6.96%	3.78e-4

Scales: per-sample source/solution; shared signed-error scale.

Quantile-selected samples show that the learned directional source split supports 2D solution reconstruction across the observed relative-error range, not only on a single favorable case.

Takeaway: Coupled Axial Green Solvers

GreenNet learns line-wise Green kernels; CouplingNet learns the directional source split.

A 2D elliptic problem is solved through axial Green inversions and a learned, balance-preserving source decomposition.

AXIAL GREEN KERNELS

Normalize axial intervals and learn line-wise inverse kernels.

ANALYTIC STRUCTURE

Encode singular Green behavior before neural correction.

SOURCE SPLIT

Learn φ/ψ without reference-solution or split labels.

ENERGY BOUND

Connect unsupervised split consistency to final solution error under assumptions.

**GreenNet supplies line-wise Green inverses;
CouplingNet learns the source split that turns them into a 2D elliptic solver.**

Backup / Q&A Menu

Details available if the audience asks

READY WITHOUT FIGURES

- Backup A: Dirac/Heaviside derivation sketch
- Backup B: imperfect Green reconstruction perturbation
- Backup C: connected-interval pull-back detail

DEFERRED UNTIL FIGURES ARE SELECTED

- extra GreenNet reconstruction evidence
- extra CouplingNet split/energy evidence

Backup A: Dirac/Heaviside Derivation Sketch

Why the analytic Green wrapping has three terms

Operator application creates two effects.

$$G_\theta = EMR_\theta + B\left(J_0 - \frac{1}{2}E\right) + AG_0.$$

EFFECT 1: DIRAC JUMP

$$\partial_t^2 G_0(t,\eta) = -\delta(t-\eta)$$

The base term (A(t)G_0(t,)) provides the required source singularity.

EFFECT 2: HEAVISIDE LEFTOVER

$$\partial_t J_0(t,\eta) = G_0(t,\eta)$$

Applying the operator to the jump structure also creates a Heaviside-type contribution.

ANALYTIC COMPENSATION

$$S = J_0 - \frac{1}{2}E, \qquad \partial_t^2 S = \partial_t G_0$$

(B(t)S(t,)) cancels the Heaviside contribution before the neural correction is used.

$$A(t) = \frac{1}{a_{\mathrm{unit}}(t)}, \qquad B(t) = \frac{a'_{\mathrm{unit}}(t) + b_{\|,\mathrm{unit}}(t)}{a_{\mathrm{unit}}(t)^2}.$$

The neural term (E(t,)M(t)R_(t,)) learns the remaining smooth residual.

Backup B: Imperfect Green Reconstruction Perturbation

What changes when the learned Green inverse is not exact?

$$G_x[\phi_*] = u_* + \varepsilon_x, \qquad G_y[\psi_*] = u_* + \varepsilon_y.$$

$$\|u_{\text{pred}} - u_*\|_a \leq \frac{C_E}{2}\left(\sqrt{\mathcal{E}_{\text{split}}} + \|\varepsilon_x - \varepsilon_y\|_a\right) + \frac{1}{2}\|\varepsilon_x + \varepsilon_y\|_a.$$

DIRECTIONAL MISMATCH

$\varepsilon_x - \varepsilon_y$

Visible to the split-consistency argument.

COMMON BIAS

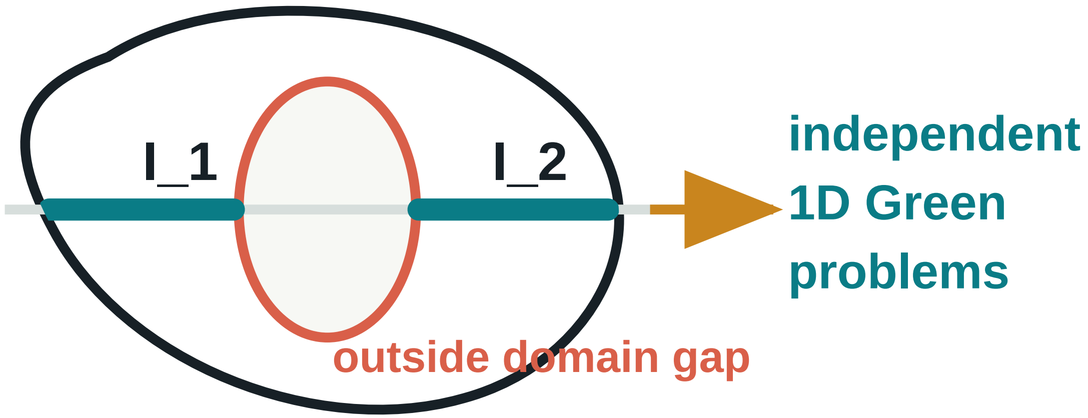
$\varepsilon_x + \varepsilon_y$

Can remain invisible when the two reconstructions agree.

Backup C: Connected-Interval Pull-Back Detail

How non-square domains become normalized one-dimensional Green problems

ONE AXIAL SLICE CAN PRODUCE MULTIPLE CONNECTED INTERVALS



PULL BACK EACH CONNECTED INTERVAL SEPARATELY

$$I = [s_0, s_1], \qquad s = s_0 + Lt, \qquad t \in [0, 1].$$

$$a_{\text{unit}} = a_{\text{phys}}, \quad a'_{\text{unit}} = La'_{\text{phys}}, \quad b_{\parallel,\text{unit}} = Lb_{\parallel,\text{phys}}, \quad c_{\text{unit}} = L^2 c_{\text{phys}}, \quad f_{\text{unit}} = L^2 f_{\text{phys}}.$$

Each connected interval is an independent one-dimensional Dirichlet Green problem.
Connected intervals are not merged across outside-domain gaps, because that would create artificial information flow outside the domain.